PHYS121 Integrated Science-Physics

W5 T1 Fluids (part 1)

References:
[1] David Halliday, Jearl Walker, Resnick Jearl, ‘Fundamentals of Physics’, (Wiley, 2018)
[2] Doug Giancoli, ‘Physics for Scientists and Engineers with modern physics’, (Pearson, 2009)
[3] Hugh D. Young, Roger A. Freedman, ‘University Physics with Modern Physics’, (Pearson, 2012)
And others specified when needed.

昆山杜克大学
DUKE KUNSHAN
UNIVERSITY

Learning Outcomes

- Perform calculations of basic properties of fluid, i.e., about density, pressure, buoyant force.
- Apply the equation of continuity of ideal fluids.
- Analyze typical examples of application of Bernoulli's principle/equation.
- Explain the phenomenon of viscosity and perform simple calculation on it.

Fluid Density and Pressure

- Physics of fluids is the basis of hydraulic engineering
- A **fluid** is a substance that can flow, like water or air, and conform to a container
- This occurs because <u>*fluids cannot sustain a shearing force*</u> (tangential to the fluid surface)
- They can however apply a force perpendicular to the fluid surface
- Some materials (pitch) take a long time to conform to a container, but are still fluids
- The essential identifier is that fluids *do not have a crystalline structure*

- The **density**, ρ, is defined as:

$$\rho = \frac{\Delta m}{\Delta V}. \qquad \textbf{Equation (14-1)}$$

- In theory the density at a point is the limit for an infinitesimal volume, but we assume a fluid sample is large relative to atomic dimensions and has uniform density. Then

$$\rho = \frac{m}{V} \qquad \textbf{Equation (14-2)}$$

- Density is a scalar quantity
- Units kg/m^3

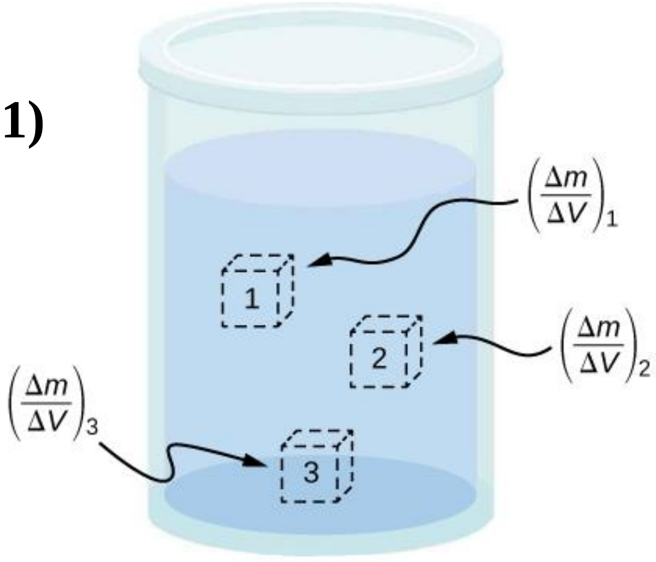


Fig. 14.4, from OpenStax textbook

- The **pressure**, force acting on an area, is defined as:

$$P = \frac{\Delta F}{\Delta A}.$$

Equation (14-3)

- We could take the limit of this for infinitesimal area, but if the force is uniform over a flat area *A* we write

$$P = \frac{F}{A}$$

Equation (14-4)

- We can measure pressure with a sensor

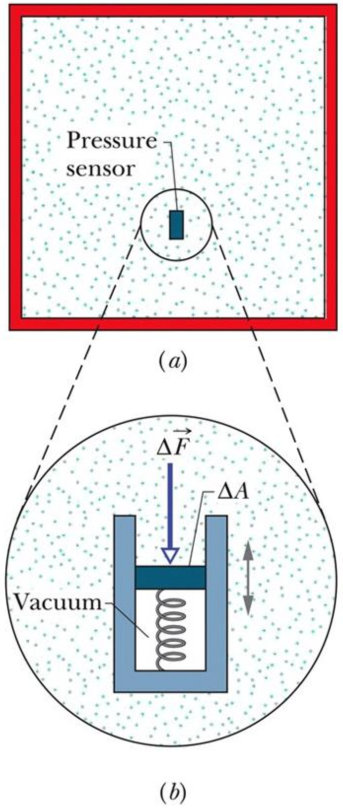
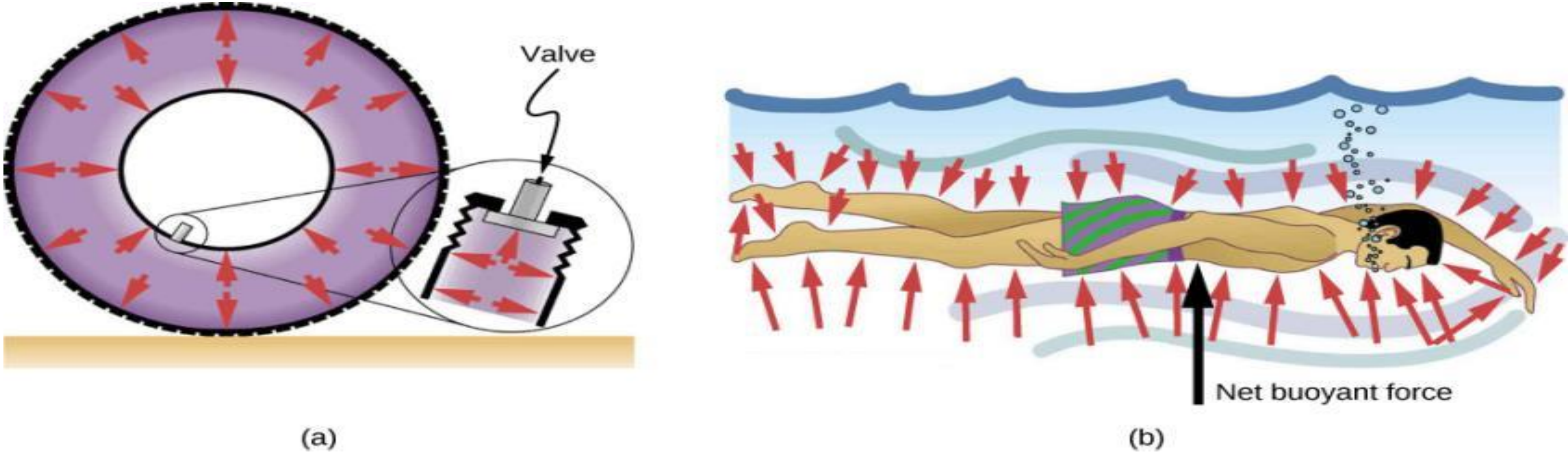


FIGURE 14.10



(a) Pressure inside this tire exerts forces perpendicular to all surfaces it contacts. The arrows represent directions and magnitudes of the forces exerted at various points.

(b) Pressure is exerted perpendicular to all sides of this swimmer, since the water would flow into the space he occupies if he were not there. The arrows represent the directions and magnitudes of the forces exerted at various points on the swimmer. Note that the forces are larger underneath, due to greater depth, giving a net upward or buoyant force. The net vertical force on the swimmer is equal to the sum of the buoyant force and the weight of the swimmer.

- We find by experiment that for a fluid at rest, *pressure has the same value at a point regardless of sensor orientation*
- Therefore static pressure is scalar, even though force is not
- Only the magnitude of the force is involved
- Units: the pascal $\left(1\ \mathrm{Pa} = 1\ \mathrm{N/m^2}\right)$

  the atmosphere (atm)

  the torr (1 torr = 1 mm Hg)

  the pound per square inch (psi)

$$1\ \mathrm{atm} = 1.01 \times 10^5 \mathrm{pa} = 760\ \mathrm{torr} = 14.7\ \mathrm{lb/in.}^2.$$

Fluids at Rest

- Hydrostatic pressures are those caused by fluids at rest (air in the atmosphere, water in a tank)

$$F_2 = F_1 + mg.$$

$$p_2 A = p_1 A + \rho A g (y_1 - y_2)$$

$$p_2 = p_1 + \rho g (y_1 - y_2).$$

$$p = p_0 + \rho g h \quad \text{(14-8)}$$

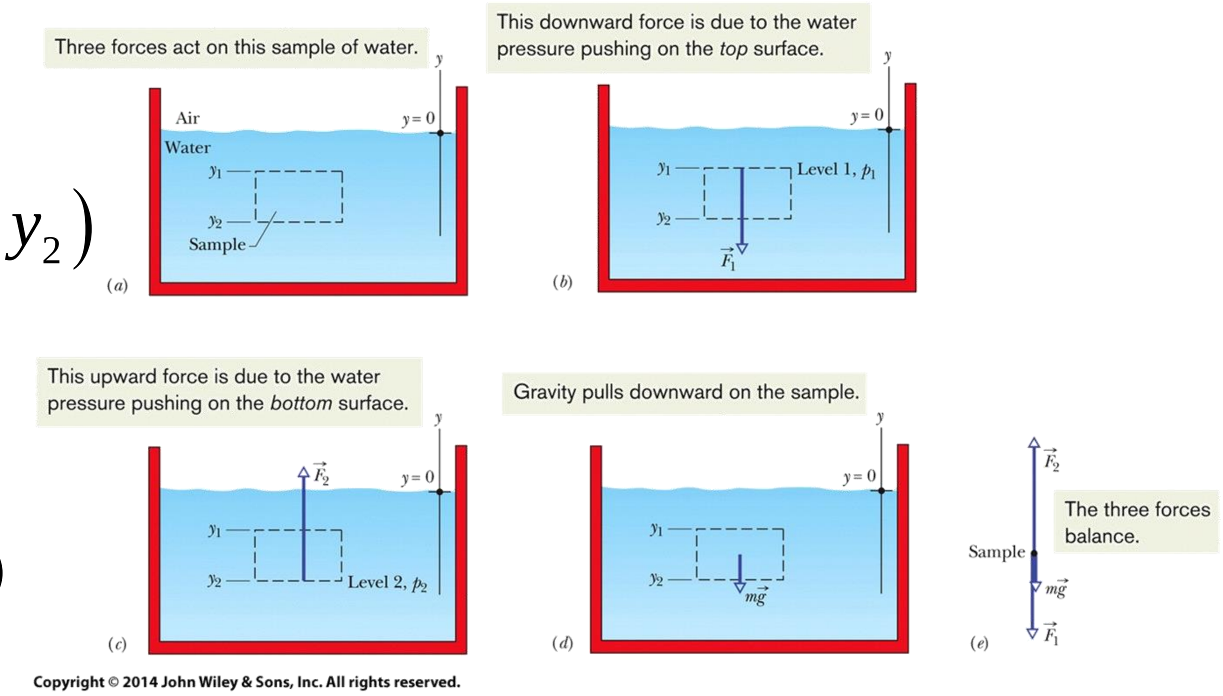


Figure 14-2

The pressure at a point in a fluid in static equilibrium depends on *the depth* of that point but not on any horizontal dimension of the fluid or its container.

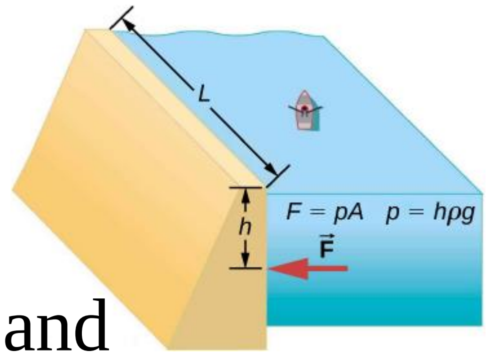


Fig. 14.1, from OpenStax textbook

- The pressure in 14-8 is the *absolute pressure*
- Consists of p_0, the pressure due to the atmosphere, and the additional pressure from the fluid
- The difference between absolute pressure and atmospheric pressure is called the ***gauge pressure*** because we use a gauge to measure this *pressure difference*
- The equation can be turned around to calculate the atmospheric pressure at a given height above ground:

$$p = p_0 - \rho_{\text{air}} g d.$$

FIGURE 14.11

(a) (b) (c)

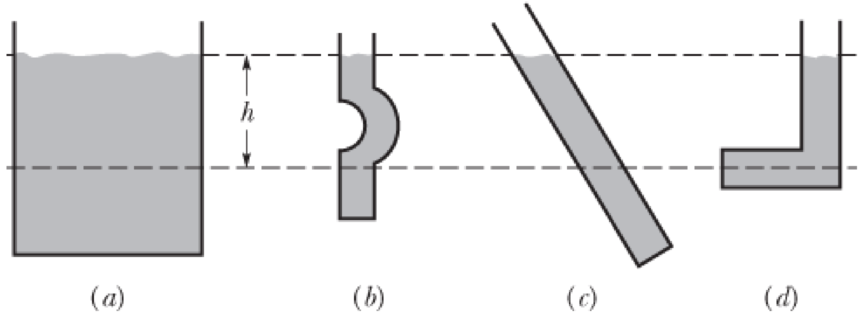
(a) Gauges are used to measure and monitor pressure in gas cylinders. Compressed gases are used in many industrial as well as medical applications.

(b) Tire pressure gauges come in many different models, but all are meant for the same purpose: to measure the internal pressure of the tire. This enables the driver to keep the tires inflated at optimal pressure for load weight and driving conditions.

(c) An ionization gauge is a high-sensitivity device used to monitor the pressure of gases in an enclosed system. Neutral gas molecules are ionized by the release of electrons, and the current is translated into a pressure reading. Ionization gauges are commonly used in industrial applications that rely on vacuum systems.

Checkpoint 1

The figure shows four containers of olive oil. Rank them according to the pressure at depth *h*, greatest first.



Answer:

All the pressures will be the same. All that matters is the distance *h*, from the surface to the location of interest, and *h* is the same in all cases.

Measuring Pressure

- Figure 14-5 shows mercury barometers
- The height difference between the air-mercury interface and the mercury level is *h*:

$$p_0 = \rho g h, \qquad \textbf{Equation (14-9)}$$

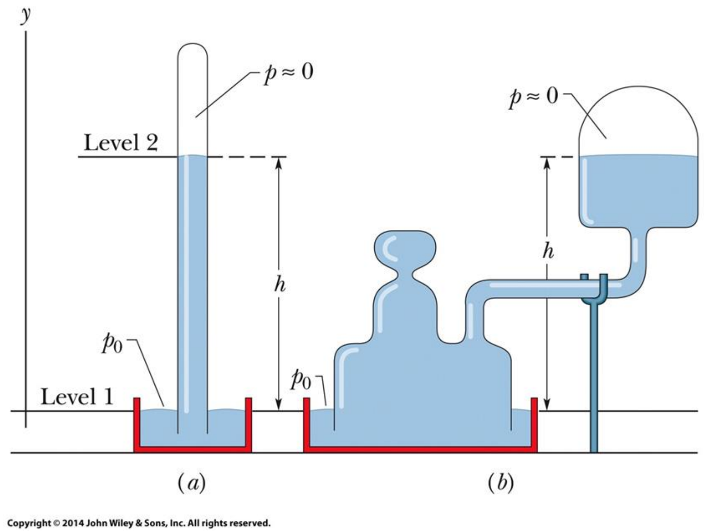


Figure 14-5

- Only the height matters, not the cross-sectional area
- Height of mercury column is numerically equal to torr pressure only if:
  - Barometer is at a place where *g* has its standard value
  - Temperature of mercury is 0°C

- The height difference between
- the two interfaces, *h*, is related
- to the gauge pressure:

$$p_g = p - p_0 = \rho g h,$$

Equation (14-10)

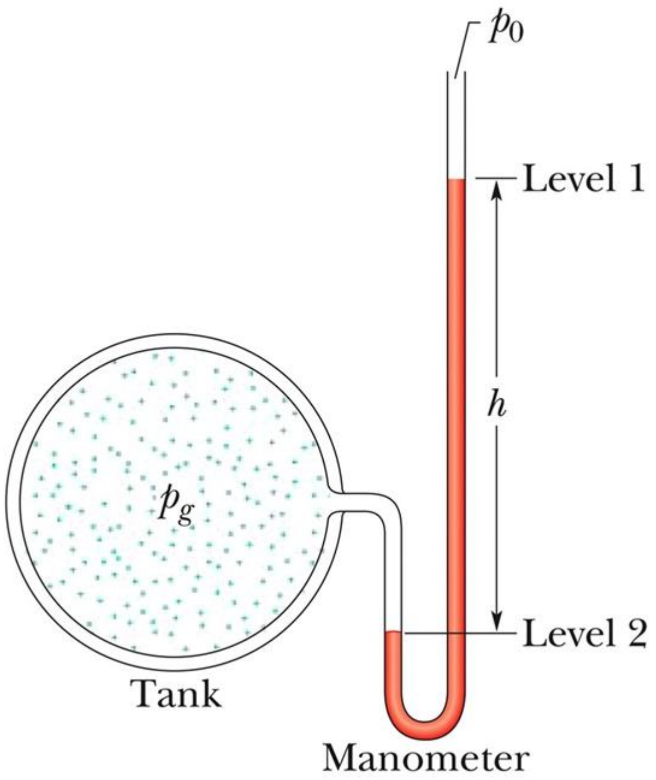
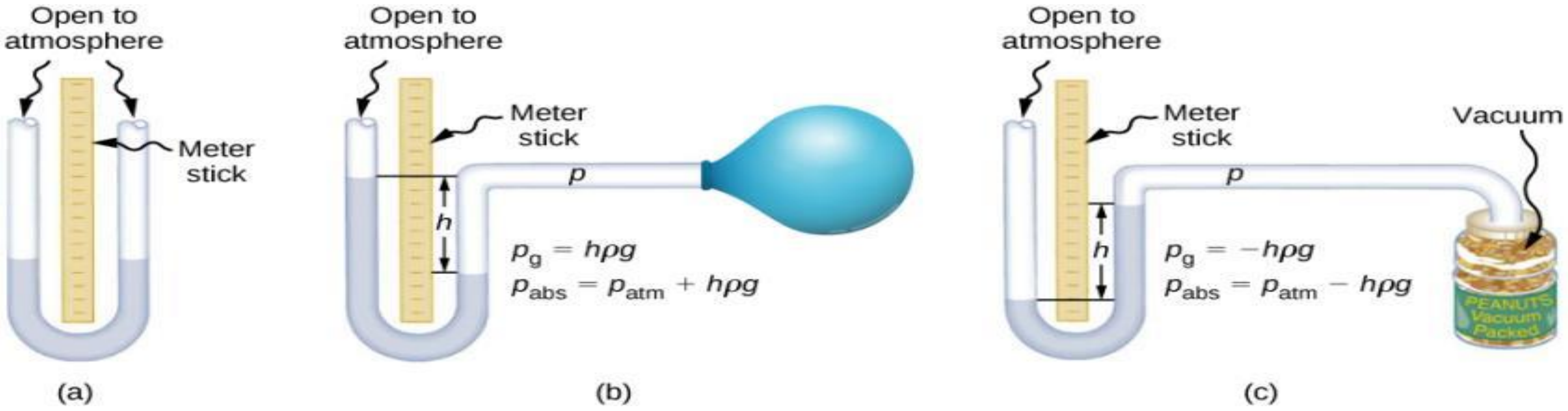


Figure 14-6 shows an open-tube manometer

- The gauge pressure can be positive or negative, depending on whether the pressure being measured is greater or less *than atmospheric pressure*

FIGURE 14.12



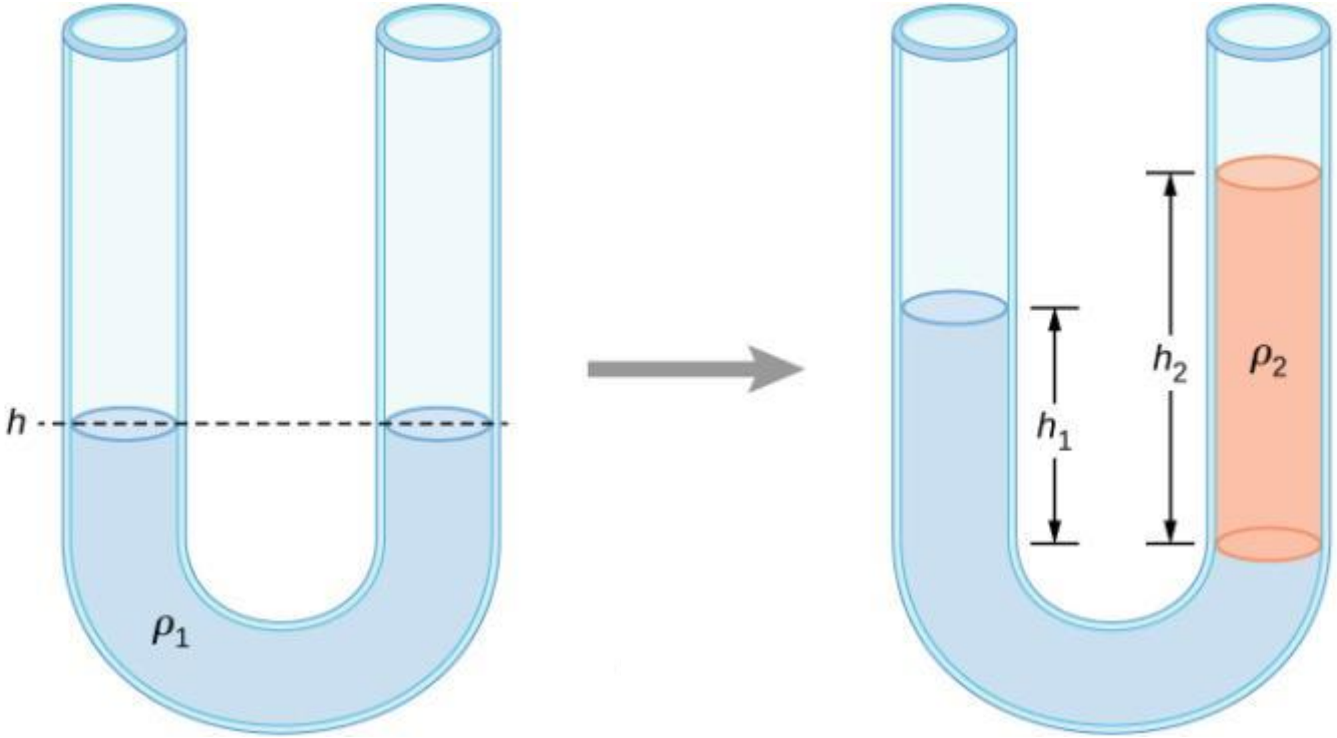
An open-tube manometer has one side open to the atmosphere.

(a) Fluid depth must be the same on both sides, or the pressure each side exerts at the bottom will be unequal and liquid will flow from the deeper side.

(b) A positive gauge pressure $p_g = h\rho g$ transmitted to one side of the manometer can support a column of fluid of height h.

(c) Similarly, atmospheric pressure is greater than a negative gauge pressure p_g by an amount $h\rho g$. The jar's rigidity prevents atmospheric pressure from being transmitted to the peanuts.

FIGURE 14.14



Two liquids of different densities are shown in a U-tube.

Which one has a higher density?

How to find the height difference?

Pascal's Principle

- **Pascal's principle** governs the transmission of pressure through an incompressible fluid:

A change in the pressure applied to an enclosed incompressible static fluid is *transmitted undiminished* to every portion of the fluid and to the walls of its container.

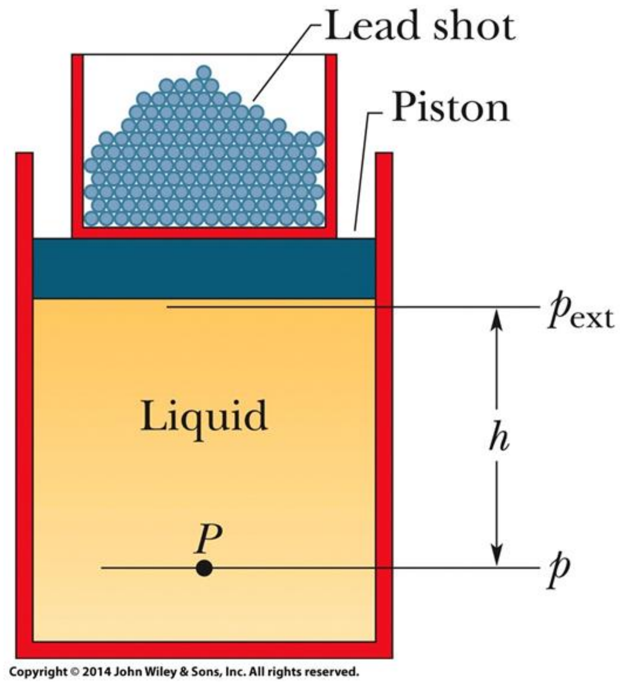


Figure 14-7

- Consider a cylinder of fluid (Figure 14-7)
- Increase p_{ext}, and p at any point must change

$$\Delta p = \Delta p_{\text{ext}}.$$

Equation (14-12)

- Independent of h

- Describes the basis for a hydraulic lever
- Input and output forces related

$$F_o = F_i \frac{A_o}{A_i}.$$

Equation (14-13)

- The distances of movement are related by:

$$d_o = d_i \frac{A_i}{A_o}.$$

Equation (14-14)

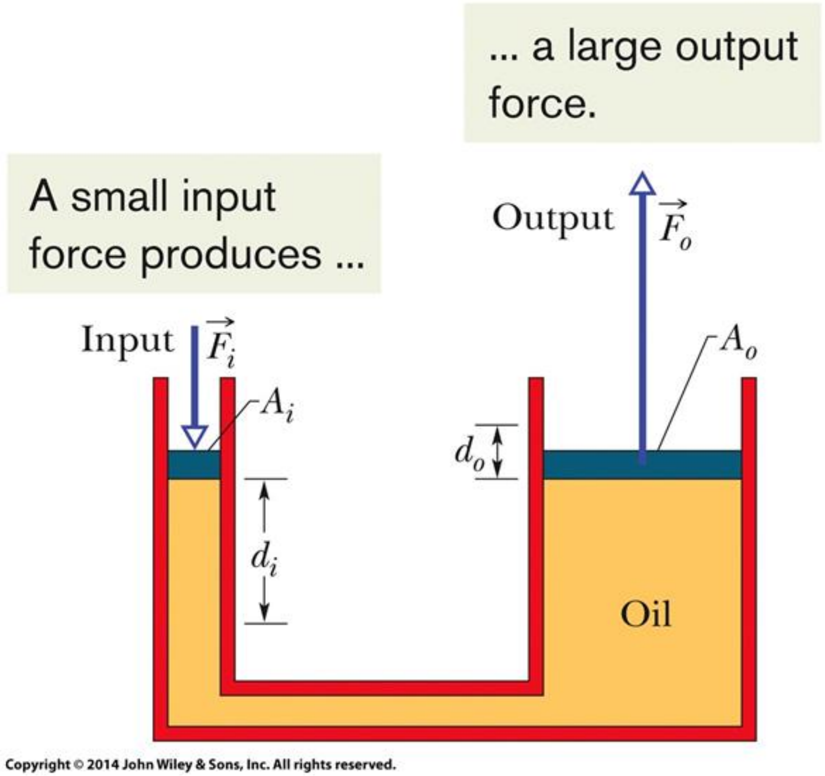


Figure 14-8

- So the work done on the input piston equals the work output

$$W = F_o d_o = \left(F_i \frac{A_o}{A_i} \right) \left(d_i \frac{A_i}{A_o} \right) = F_i d_i, \qquad \textbf{Equation (14-15)}$$

- The advantage of the hydraulic lever is that:

With a hydraulic lever, a given force applied over a given distance can be transformed to a greater force applied over a smaller distance.

FIGURE 14.17

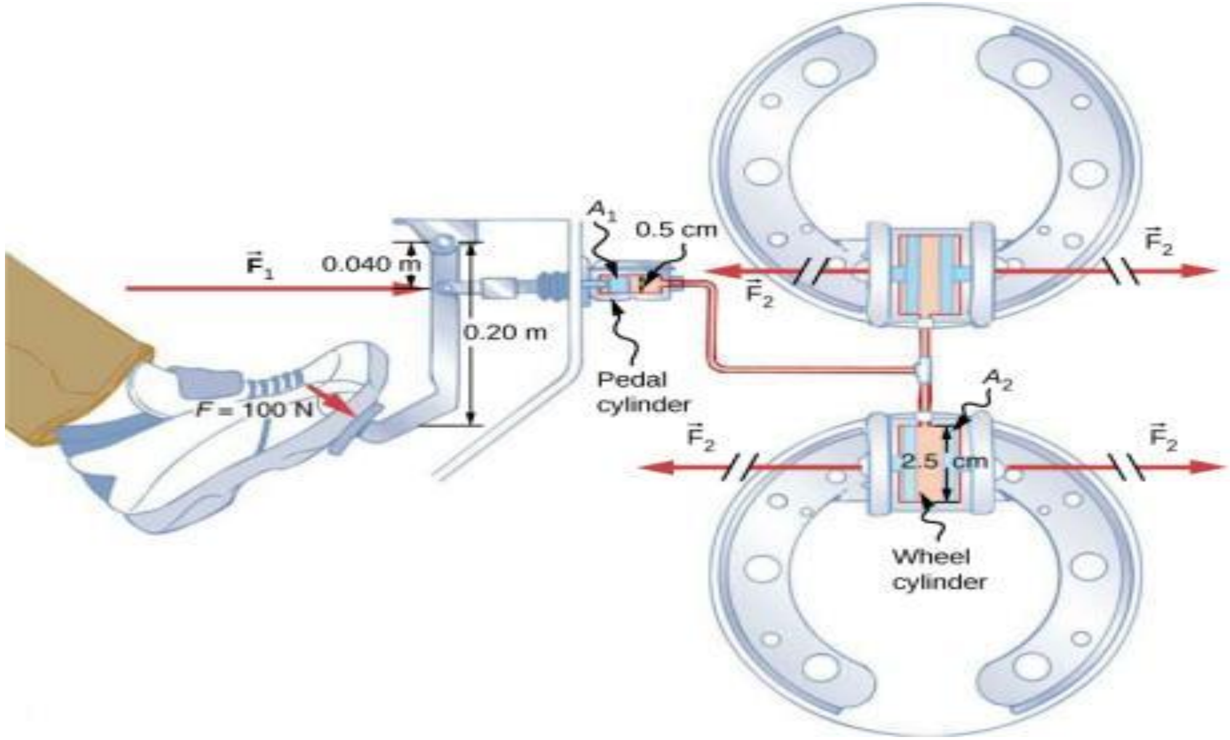
(a)

(b)

(a) A hydraulic jack operates by applying forces (F_1 , F_2) to an incompressible fluid in a U-tube, using a movable piston (A_1, A_2) on each side of the tube.

(b) Hydraulic jacks are commonly used by car mechanics to lift vehicles so that repairs and maintenance can be performed.

FIGURE 14.18



Hydraulic brakes use Pascal's principle. The driver pushes the brake pedal, exerting a force that is increased by the simple lever and again by the hydraulic system. Each of the identical wheel cylinders receives the same pressure and, therefore, creates the same force output F_2. The circular cross-sectional areas of the pedal and wheel cylinders are represented by A_1 and A_2, respectively.

How large if F_2?

How about the work done by F_1 and F_2?

Archimedes' Principle

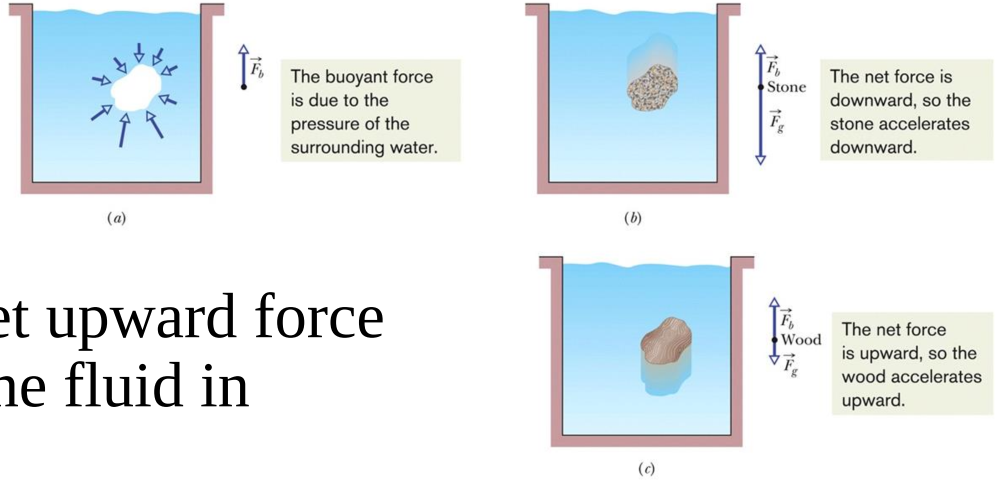


Figure 14-10

- The **buoyant force** is the net upward force **on** a submerged object **by** the fluid in which it is submerged
- This force opposes the weight of the object

  It comes from the increase in pressure with depth
- The stone and piece of wood *displace the water that would otherwise occupy that space*

- **Archimedes' Principle** states that:

When a body is fully or partially submerged in a fluid, a buoyant force $\vec{F}_b$ surrounding fluid acts on the body. The force is directed upward and has a magnitude equal to the weight $m_f\, g$ of the fluid that has been displaced by the body.

- The buoyant force has magnitude

$$F_b = m_f\, g$$

Equation (14-16)

where m_f is the mass of displaced fluid

- A block of wood in static equilibrium is *floating*:

When a body floats in a fluid, the magnitude F_b of the buoyant force on the body is equal to the magnitude F_g of the gravitational force on the body.

- This is expressed: $F_b = F_g \quad (\text{floating}).$ **Equation (14-17)**

- Because of Equation 14-16 we know:

When a body floats in a fluid, the magnitude F_g of the gravitational force on the body is equal to the weight $m_f g$ of the fluid that has been displaced by the body

- Which means: $F_g = m_f g \quad (\text{floating}).$ **Equation (14-18)**

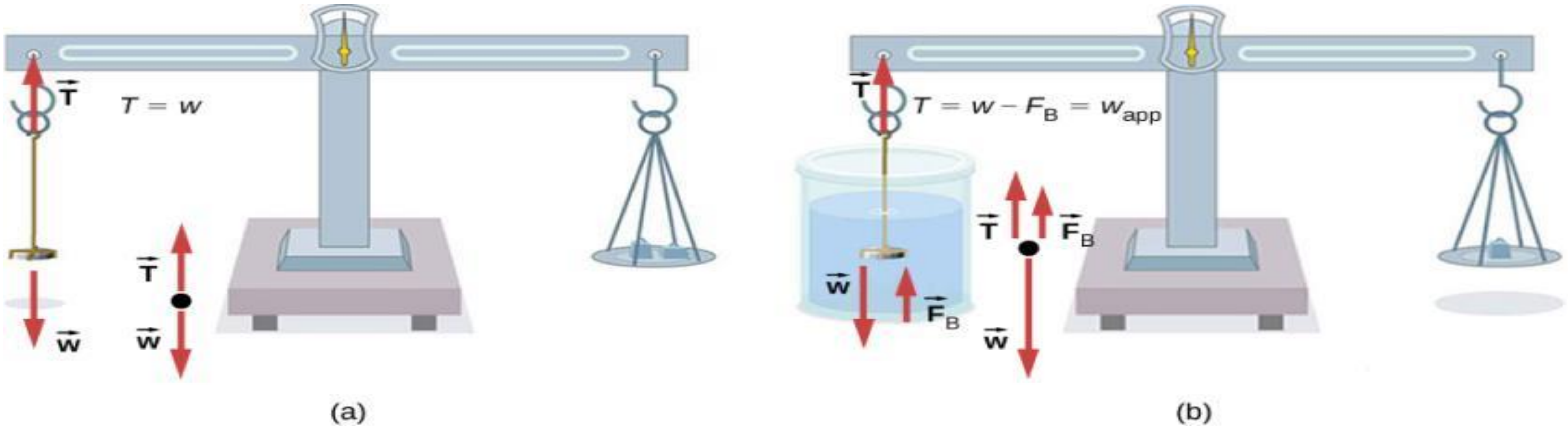
- The *apparent weight* of a body in a fluid is related to the actual weight of the body by:

$$(\text{apparent weight}) = (\text{actual weight}) - (\text{buoyant force})$$

- We write this as:

$$\text{weight}_{\text{app}} = \text{weigth} - F_b \quad (\text{apparent weight}).$$

FIGURE 14.23



(a) A coin is weighed in air.

(b) The apparent weight of the coin is determined while it is completely submerged in a fluid of known density. These two measurements are used to calculate the density of the coin.

Checkpoint 2

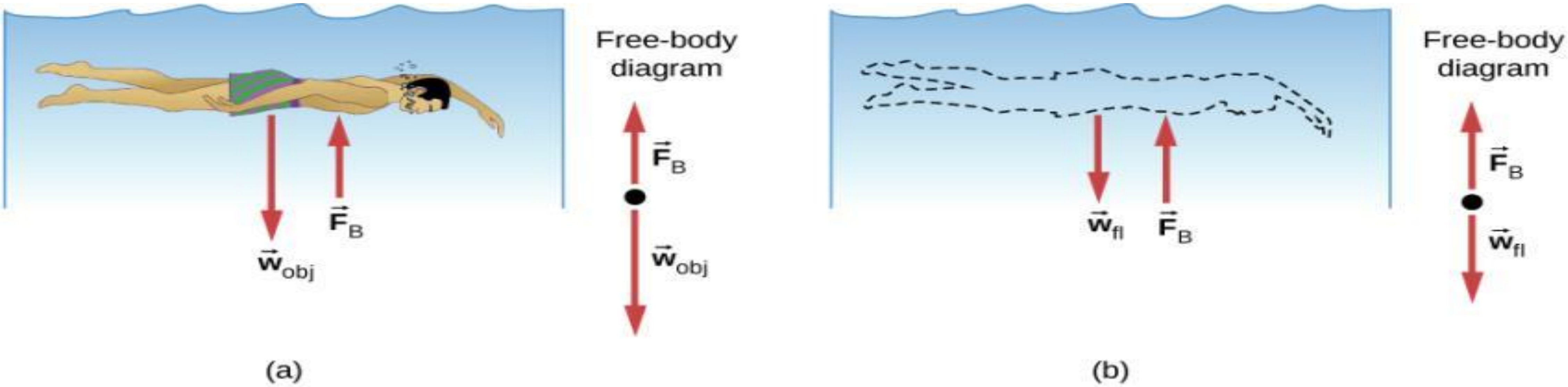
A penguin floats first in a fluid of density ρ_0, then in a fluid of density $0.95\rho_0$, and then in a fluid of density $1.1\rho_0$. (a) Rank the densities according to the magnitude of the buoyant force on the penguin, greatest first. (b) Rank the densities according to the amount of fluid displaced by the penguin, greatest first.

Answer:

(a) all the same

(b) $0.95\rho_0, 1\rho_0, 1.1\rho_0$

FIGURE 14.21



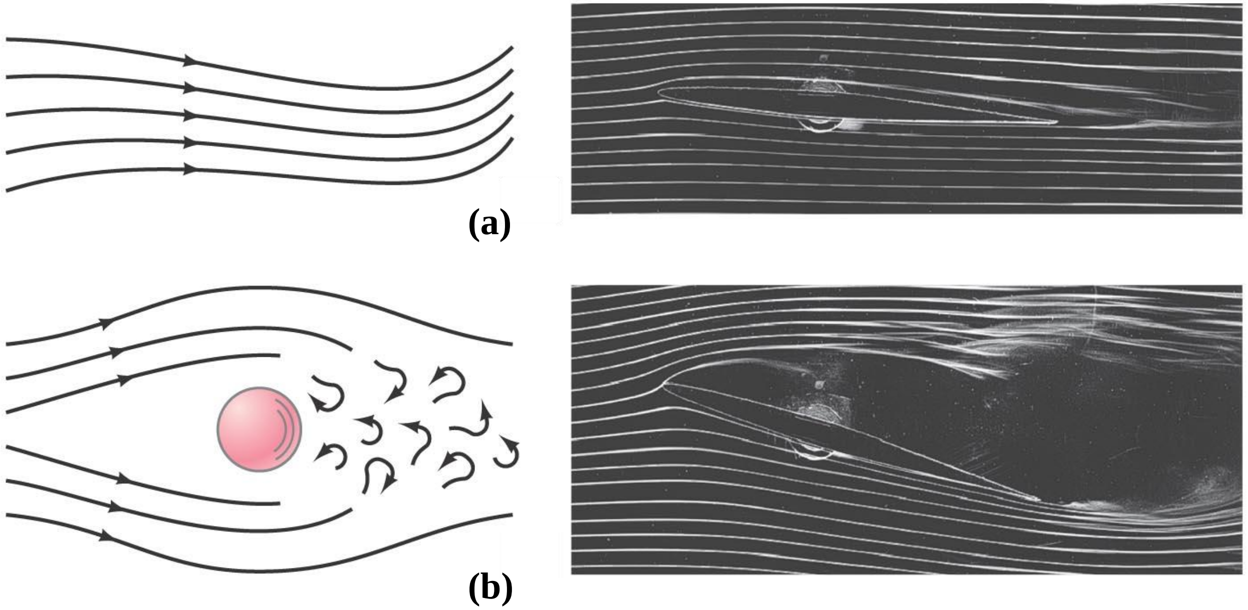
(a) An object submerged in a fluid experiences a buoyant force F_B. If F_B is greater than the weight of the object, the object rises. If F_B is less than the weight of the object, the object sinks.

(b) If the object is removed, it is replaced by fluid having weight w_{fl}. Since this weight is supported by surrounding fluid, the buoyant force must equal the weight of the fluid displaced.

Fluids in Motion; Flow Rate and the Equation of Continuity

If the flow of a fluid is smooth, it is called **streamline** or **laminar** flow (a).

Above a certain speed, the flow becomes **turbulent** (b). Turbulent flow has **eddies**; the **viscosity** of the fluid is much greater when eddies are present.



We will deal with **laminar** flow.

The **mass flow rate** is the mass that passes a given point per unit time. The flow rates at any two points must be **equal**, as long as no fluid is being added or taken away.

This gives us the **equation of continuity**:

Since $$\frac{\Delta m_1}{\Delta t} = \frac{\Delta m_2}{\Delta t},$$

then $$\rho_1 A_1 v_1 = \rho_2 A_2 v_2 .$$

If the density doesn't change—typical for liquids—this simplifies to $A_1 v_1 = A_2 v_2$. Where the pipe is wider, the flow is slower.

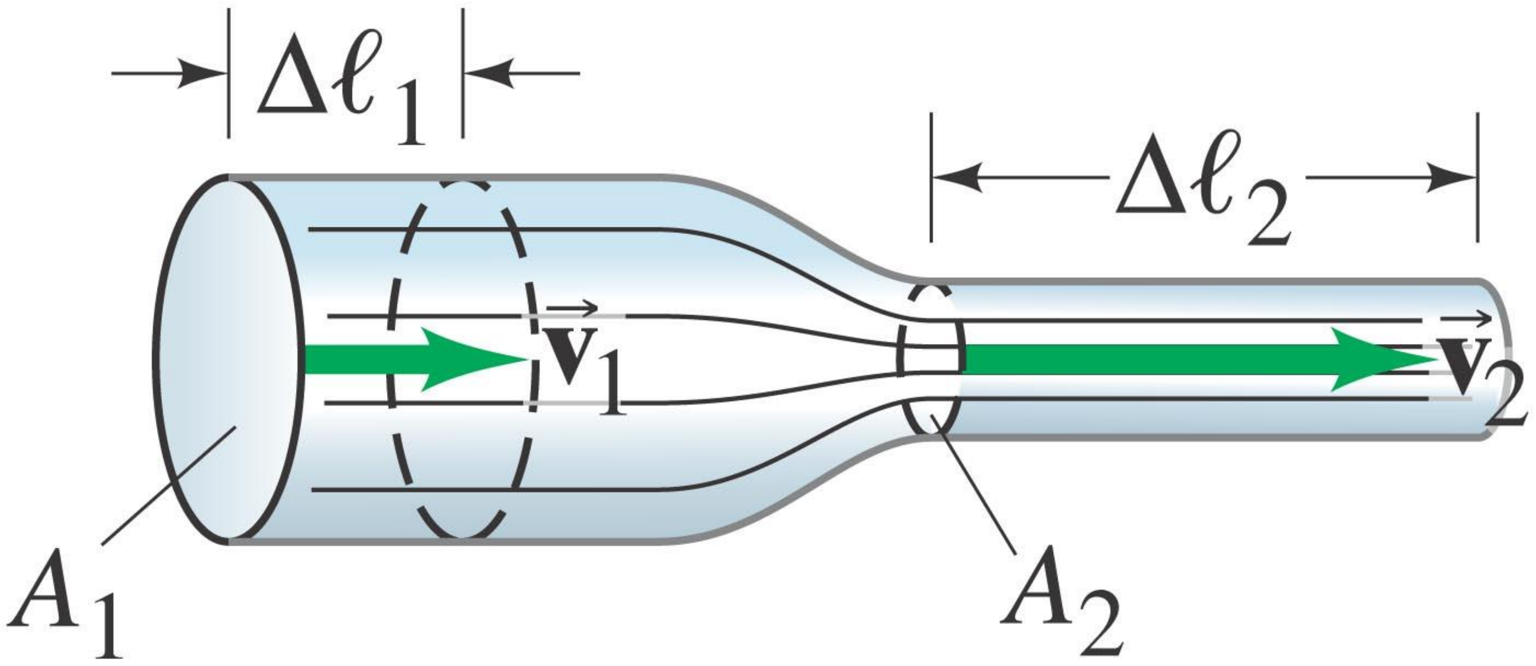
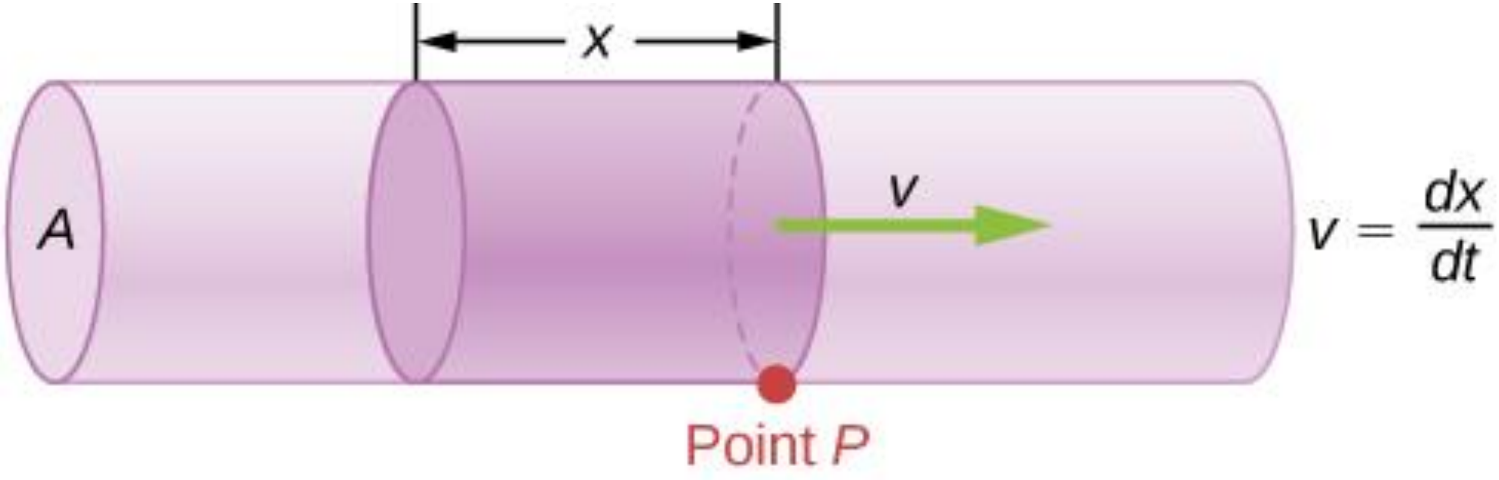


FIGURE 14.26



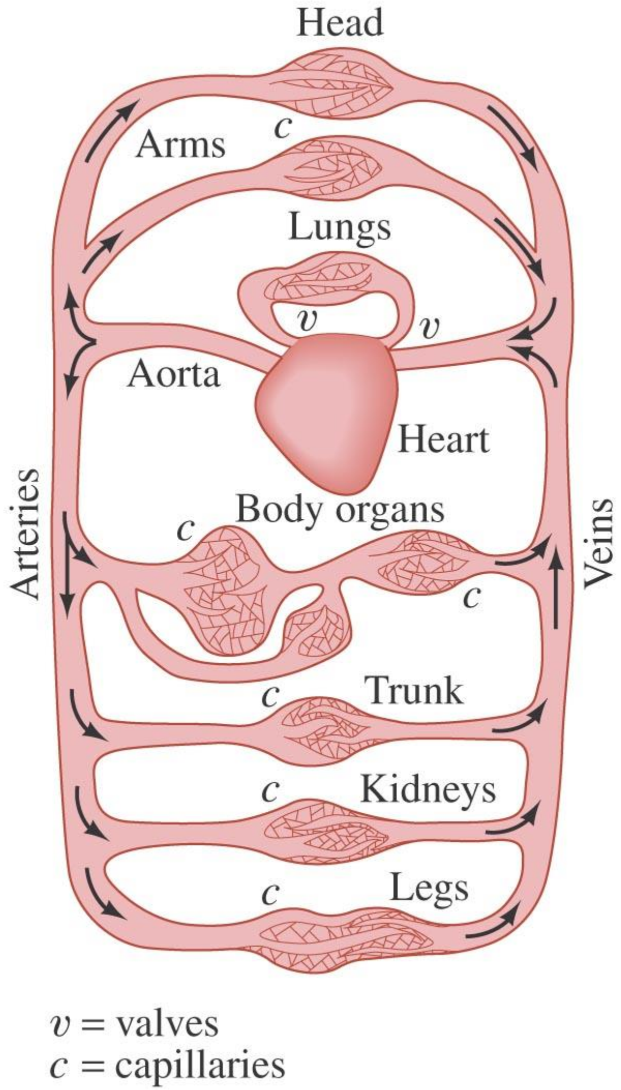
$$Q = \frac{dV}{dt} = \frac{d}{dt}(Ax) = A\frac{dx}{dt} = Av$$

Flow rate is the volume of fluid flowing past a point through the area *A* per unit time. Here, the shaded cylinder of fluid flows past point *P* in a uniform pipe in time *t*.

Since liquids are essentially incompressible, the equation of continuity is valid for all liquids. However, gases are compressible, so the equation must be applied with caution to gases if they are subjected to compression or expansion.

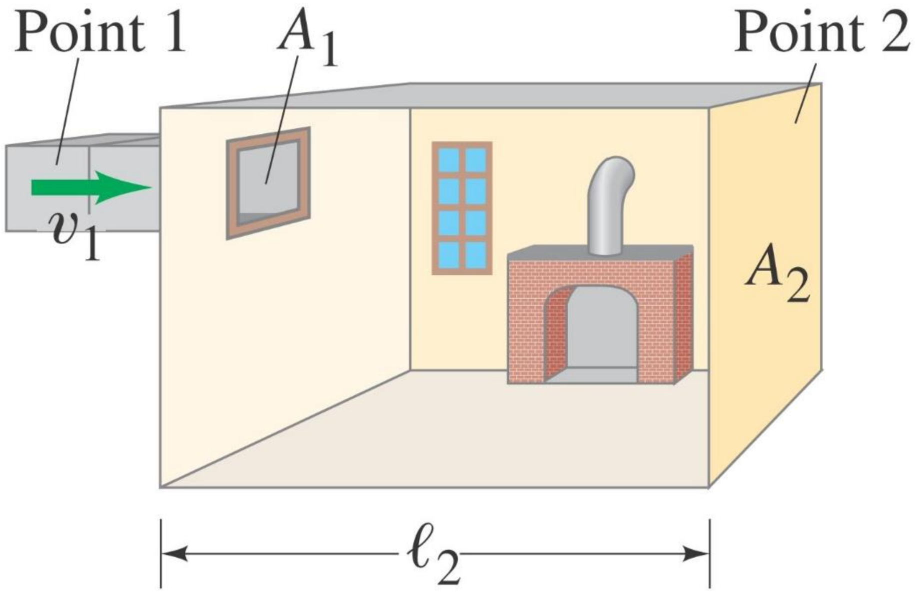
Example: Blood flow.

In humans, blood flows from the heart into the aorta, from which it passes into the major arteries. These branch into the small arteries (arterioles), which in turn branch into myriads of tiny capillaries. The blood returns to the heart via the veins. The radius of the aorta is about 1.2 cm, and the blood passing through it has a speed of about 40 cm/s. A typical capillary has a radius of about 4×10^{-4} cm, and blood flows through it at a speed of about 5×10^{-4} m/s. Estimate the number of capillaries that are in the body.



Example: Heating duct to a room.

What area must a heating duct have if air moving 3.0 m/s along it can replenish the air every 15 minutes in a room of volume 300 m^3? Assume the air's density remains constant.



Learning Outcomes

- Perform calculations of basic properties of fluid, i.e., about density, pressure, buoyant force.
- Apply the equation of continuity of ideal fluids.
- Analyze typical examples of application of Bernoulli's principle/equation.
- Explain the phenomenon of viscosity and perform simple calculation on it.

昆山杜克大学
DUKE KUNSHAN UNIVERSITY

CHAPTER REVIEW AND EXAMPLES, DEMOS

https://openstax.org/books/university-physics-volume-1/pages/14-summary

Specific gravity is defined as the ratio of the density of the material to the density of water at 4.0°C and one atmosphere of pressure, which is 1000kg/m^3.

Note that although force is a vector, pressure is a scalar. Pressure is a scalar quantity because it is defined to be proportional to the magnitude of the force acting perpendicular to the surface area.

$$\frac{dp}{dy} = -\rho g. \tag{14.8}$$

in a uniform gravitational field.

Variation of atmospheric pressure with height

… we have assumed both a constant temperature and a constant *g* over such great distances from Earth, neither of which is correct in reality.

$$pV = Nk_BT$$

$$\frac{dp}{dy} = -p\left(\frac{mg}{k_B T}\right),$$

$$\frac{dp}{dy} = -\alpha p$$

$$\frac{dp}{p} = -\alpha dy$$

$$\int_{p_0}^{p(y)} \frac{dp}{p} = \int_0^y -\alpha dy$$

$$[\ln(p)]_{p_0}^{p(y)} = [-\alpha y]_0^y$$

$$\ln(p) - \ln(p_0) = -\alpha y$$

$$\ln\left(\frac{p}{p_0}\right) = -\alpha y$$

Questions